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Correction of the Physical Sensor and Rational Function Models Using the Example of the SkySat SSC13 Platform

by

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in the
Institute of Computer Science and Computational Mathematics
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"Everything is related to everything else, but near things are more related than distant things."

Waldo Tobler

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Abstract

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This thesis presents a general framework for improving the geopositioning accuracy of both the Physical Sensor Model (PSM) and the Rational Function Model (RFM). The proposed approach is demonstrated using imagery acquired by the SkySat SSC13 satellite platform, and its effectiveness is evaluated using Ground Control Points (GCPs).

In addition to the correction methodology, the thesis provides a theoretical introduction to the pinhole camera model, coordinate transformations, and rotation parameterizations. Although the necessary mathematical background is discussed, the primary focus of the work is the practical application of these concepts to satellite image georeferencing.

Acknowledgements

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Their contributions were invaluable, and this work would not have been possible without their support.

Contents

Symbols

Id	Identity transformation $Id : \mathbb{R}^3 \ni x \mapsto x \in \mathbb{R}^3$
μ_L	Lebesgue's measure for $\mathbb{R}^n$ vector spaces
$\backsim$	equivalence relation defined by $x \backsim y :\Leftrightarrow x = -y$
$\mathbb{D}^n_\pi$	n-dimensional disk with radius of π : $\{x \in \mathbb{R}^n \mid \|\|x\|\| \leq \pi\}$
$\mathbf{S}^n$	n-dimensional unit sphere : $\{x \in \mathbb{R}^{n+1} \mid \|\|x\|\| = 1\}$
$\mathbb{S}^n$	n-dimensional unit sphere : $\mathbb{R}^n / \mathbb{Z}^n$
$\mathbb{T}^n$	n-dimensional unit torus : $(\mathbb{S}^1)^n = \mathbb{S}^1 \times \mathbb{S}^1 \times \ldots \times \mathbb{S}^1$

Chapter 1

Introduction

Geopositioning has become one of the most important technologies of the last three decades. Many modern services, such as Google Earth, rely on fast and accurate estimation of the geographic location of objects and individuals. It is also of critical importance in military applications. Unfortunately, Global Navigation Satellite Systems (GNSS), such as GPS and Galileo, cannot operate without ground-based receivers. As a result, they become ineffective in inaccessible regions due to political restrictions, remoteness, or lack of infrastructure. Furthermore, receivers must be deployed at specific locations to collect measurements, which significantly limits the range of possible applications. For these reasons, satellite imagery and orthorectification play a crucial role in modern geopositioning systems.

The use of satellite imagery for geopositioning requires a model capable of transforming points on the Earth's surface into pixel coordinates in an image. By inverting such a model, it becomes possible to estimate an object's geographic location solely from its appearance in the image. Two commonly used models for this purpose are the Physical Sensor Model (PSM) and the Rational Function Model (RFM).

The Physical Sensor Model provides detailed information about the satellite's sensors together with a camera model. This information can be used to transform a point expressed in the global reference frame (ECEF in this work) into the satellite reference frame and subsequently into the payload (camera) reference frame. The camera model is then used to determine the corresponding position of the point in the image plane. Despite its many advantages, the PSM has one significant drawback: it exposes sensitive information about the satellite, including its position and orientation. Consequently, its use is often restricted, and satellite operators rarely make such data publicly available.

An alternative approach is the Rational Function Model. Similar to the PSM, it describes the transformation from the global reference frame to image coordinates. However, instead of explicitly providing sensor and platform parameters, all relevant information is encoded within a set of rational functions. As a result, the exact satellite position and orientation cannot be readily recovered. Both models are highly accurate, with the PSM generally achieving slightly better performance. Furthermore, both approaches can be refined to improve their accuracy. Nevertheless, the PSM is considerably easier to invert, which is an important advantage in geopositioning applications.

Despite their accuracy, both models are sensitive to sensor misalignment, numerical errors, and other inaccuracies introduced during model derivation. For example, an error of only 0.001% in the estimated rotation angle may result in a positioning error exceeding ten meters for satellites operating in low Earth orbit (LEO). Consequently, significant research effort has been devoted to improving the accuracy of both models.

The most common approach involves correcting the sensor data provided to the PSM, such as satellite position and orientation, or directly correcting the RFM itself. This is achieved by comparing points of known location, referred to as Ground Control Points (GCPs), with their positions predicted by the model. The discrepancies between the observed and predicted locations are then used to estimate correction parameters and update the model accordingly. This procedure can improve accuracy only if the systematic bias between the true and measured data can be approximated by a relatively simple function. In other words, it is assumed that random noise constitutes a smaller component of the total error than systematic effects, such as sensor misalignment introduced during satellite deployment.

1.1 Coordinate Systems

We consider four distinct frames of reference: the global frame, the satellite frame, the payload frame, and the image frame.

In our case, the global frame is the Earth-Centered, Earth-Fixed (ECEF) coordinate system, a three-dimensional Euclidean space described by Cartesian coordinates, with its origin at the Earth's center and rotating together with the Earth. This is a practical choice due to its direct relationship with geodetic coordinates.

The satellite and payload frames are also three-dimensional coordinate systems, centered at the satellite and the camera, respectively. The distinction between them is crucial because, although their positions relative to the ECEF frame are usually very similar, their orientations may differ significantly. As discussed in the introduction, even small orientation errors can lead to substantial inaccuracies in geolocation.

Finally, the image frame is a two-dimensional coordinate system whose origin is located at the upper-left corner of the image. Each integer-valued coordinate corresponds to a pixel location. Naturally, some projected points may lie between pixel centers or even outside the image boundaries. However, this should be regarded as an advantage rather than a limitation, as it allows continuous geometric transformations to be represented without artificial discretization.

1.2 Camera Model

A camera model provides a mathematical description of how a camera maps three-dimensional points to two-dimensional image coordinates. Since different cameras may have different optical properties and lens configurations, many camera models have been proposed. However, for the purposes of this work, we restrict our attention to the classical pinhole camera model.

The pinhole camera model can be viewed as a partial function

$$F_c : \mathbb{R}^3 \nrightarrow \mathbb{R}^2, \tag{1.1}$$

which maps points expressed in the camera frame (the payload frame in our case) to image coordinates. Geometrically, the mapping is obtained by projecting points onto a two-dimensional affine plane embedded in ($\mathbb{R}^3$) and then expressing the result in the image coordinate system. This construction closely resembles the physical path of light traveling from an object through a pinhole onto the image plane.

The function is necessarily partial because projection is undefined for points lying in a plane parallel to the image plane and passing through the camera center. The mapping can be written as

$$(p_x, p_y) = F_c(X, Y, Z) = \left(\frac{f_x X}{Z} + c_x, , \frac{f_y Y}{Z} + c_y \right), \tag{1.2}$$

where f_x and f_y denote the focal lengths expressed in pixel units, and c_x, c_y represent the coordinates of the principal point, i.e., the offset between the image coordinate system and the optical axis. p_x, p_y are pixel coordinates in the image frame.

For $Z = 0$, the function is left undefined, since such points cannot be projected onto the image plane.

Although F_c is not a linear transformation, it admits a convenient representation using homogeneous coordinates. The camera matrix is given by

$$\mathrm{P}_c = \begin{bmatrix} f_x & 0 & c_x & 0 \\ 0 & f_y & c_y & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}. \tag{1.3}$$

For a point (X, Y, Z), represented in homogeneous coordinates as $(X, Y, Z, 1)$, we have

$$\mathrm{P}_c \begin{pmatrix} X \\ Y \\ Z \\ 1 \end{pmatrix} = \begin{pmatrix} Zp_x \\ Zp_y \\ Z \end{pmatrix}, \tag{1.4}$$

This representation allows the perspective projection to be expressed as a linear transformation in homogeneous coordinates followed by a normalization step. As will be shown in the following sections, this formulation greatly simplifies the composition of multiple geometric transformations.

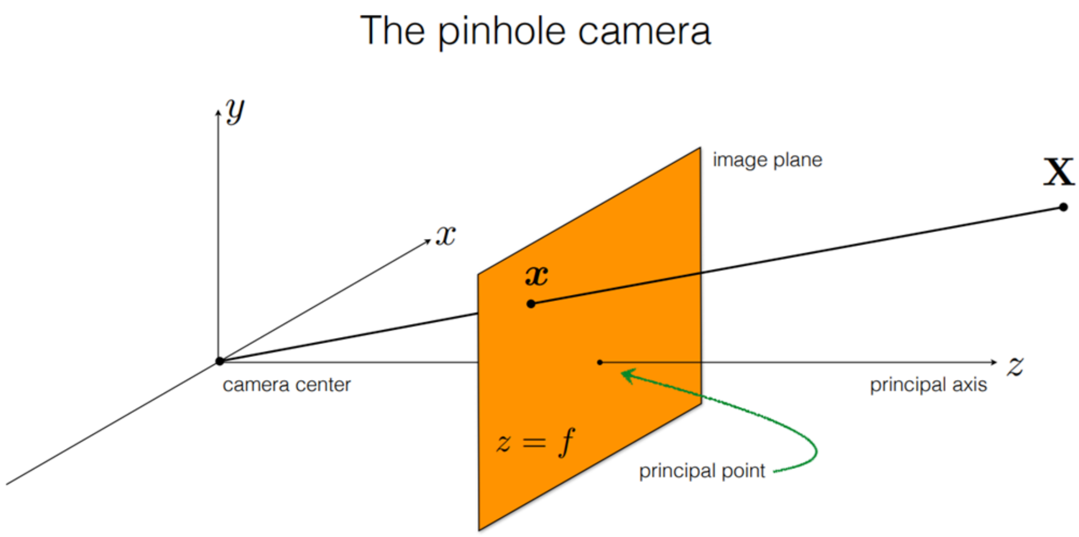

FIGURE 1.1: Visual interpretation of the camera model embedded in the camera frame (Kitani [1])

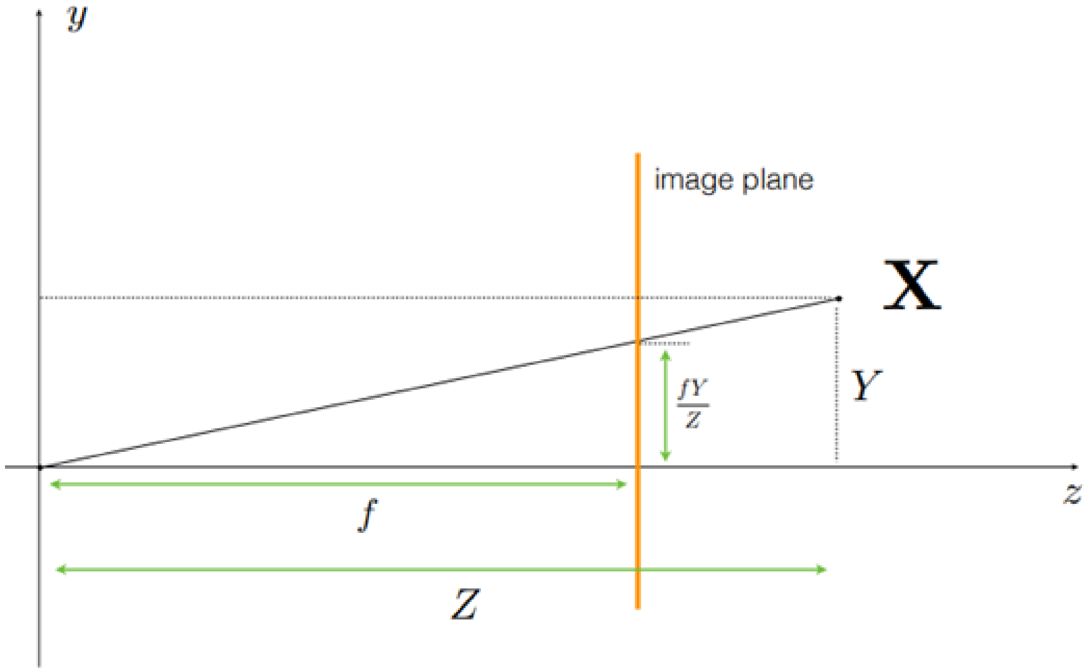

FIGURE 1.2: Visual interpretation of the function F_c on a single image axis (Kitani [1])

1.3 Translation and Rotation Between Frames

To apply the camera model, we must first determine the position and orientation of the camera with respect to the global frame. This is necessary because the domain of F_c consists of points expressed in the camera coordinate system, whereas the points of interest are typically given in global coordinates.

The translation from the global frame to the camera frame can be described by the affine transformation

$$T(x, y, z) = (x - c_x,\ y - c_y,\ z - c_z), \tag{1.5}$$

where (c_x, c_y, c_z) denotes the position of the camera center in the global frame.

Since T is an affine rather than a linear transformation, it is convenient to represent it using homogeneous coordinates. The corresponding transformation matrix is

$$\mathrm{P}_t = \begin{bmatrix} 1 & 0 & 0 & -c_x \\ 0 & 1 & 0 & -c_y \\ 0 & 0 & 1 & -c_z \\ 0 & 0 & 0 & 1 \end{bmatrix}. \tag{1.6}$$

Then, for every point (x, y, z),

$$\begin{pmatrix} T(x,y,z) \\ 1 \end{pmatrix} = \mathrm{P}_t \begin{pmatrix} x \\ y \\ z \\ 1 \end{pmatrix}. \tag{1.7}$$

After translating the coordinate system, we must also align its orientation with that of the camera frame. This is accomplished using a rotation matrix R, which can be obtained from a unit quaternion

$$\mathbf{q} = (q_w, q_x, q_y, q_z). \tag{1.8}$$

The corresponding rotation matrix is

$$R = \begin{bmatrix} 1 - 2(q_y^2 + q_z^2) & 2(q_x q_y - q_w q_z) & 2(q_w q_y + q_x q_z) \\ 2(q_x q_y + q_w q_z) & 1 - 2(q_x^2 + q_z^2) & 2(q_y q_z - q_w q_x) \\ 2(q_x q_z - q_w q_y) & 2(q_w q_x + q_y q_z) & 1 - 2(q_x^2 + q_y^2) \end{bmatrix}. \tag{1.9}$$

Although rotations are already linear transformations, we again embed the matrix in homogeneous coordinates in order to maintain a consistent representation of all geometric transformations:

$$P_r = \begin{bmatrix} 1 - 2(q_y^2 + q_z^2) & 2(q_x q_y - q_w q_z) & 2(q_w q_y + q_x q_z) & 0 \\ 2(q_x q_y + q_w q_z) & 1 - 2(q_x^2 + q_z^2) & 2(q_y q_z - q_w q_x) & 0 \\ 2(q_x q_z - q_w q_y) & 2(q_w q_x + q_y q_z) & 1 - 2(q_x^2 + q_y^2) & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}. \tag{1.10}$$

The motivation for using quaternions to parameterize rotations is discussed in Appendix A.

1.4 Physical Sensor Model (PSM)

We can now define the Physical Sensor Model (PSM). As discussed in the introduction, the PSM is a mapping from points expressed in the global frame to pixel coordinates in the image frame. It can be written as

$$(p_x, p_y) = F_c \circ R_{in} \circ T_{in} \circ R_{ex} \circ T_{ex}(X, Y, Z), \tag{1.11}$$

where F_c denotes the camera model, T_{ex} and R_{ex} represent the translation and rotation between the global and satellite frames, and T_{in} and R_{in} describe the translation and rotation between the satellite frame and the payload (camera) frame.

In practice, the model is usually expressed in homogeneous coordinates. In this representation,

$$\begin{pmatrix} Zp_x \\ Zp_y \\ Z \end{pmatrix} = \mathrm{P}_c \mathrm{P}_{in} \mathrm{P}_{ex} \begin{pmatrix} X \\ Y \\ Z \\ 1 \end{pmatrix}, \tag{1.12}$$

where P_c is the camera matrix, while P_{in} and P_{ex} are the homogeneous transformation matrices corresponding to the intrinsic and extrinsic transformations, respectively.

Consequently, evaluating the PSM requires knowledge of the satellite position, the satellite orientation, the relative position of the camera with respect to the satellite, and the camera calibration parameters.

1.5 Rational Function Model (RFM)

In contrast to the PSM, the Rational Function Model (RFM) provides the same mapping between ground coordinates and image coordinates while concealing the underlying sensor geometry. This property makes the model particularly suitable for distributing commercial satellite imagery without revealing sensitive information about the satellite platform.

The RFM is defined by a pair of rational functions

$$(P, L, H) \mapsto (l, s) = \left(\frac{N_x(B, L, H)}{D_x(B, L, H)}, \frac{N_y(B, L, H)}{D_y(B, L, H)} \right), \tag{1.13}$$

where N_x, D_x, N_y, and D_y are multivariate polynomials.

The variables B, L, and H denote normalized latitude, longitude, and altitude, respectively:

$$B = \frac{\phi - LAT\_OFF}{LAT\_SCALE}, \qquad L = \frac{\lambda - LONG\_OFF}{LONG\_SCALE},$$

$$H = \frac{h - ALT\_OFF}{ALT\_SCALE}.$$

The outputs l and s are normalized image coordinates. The corresponding pixel coordinates are obtained by

$$p_y = l \cdot LINE\_SCALE + LINE\_OFF,$$

$$p_x = s \cdot SAMPLE\_SCALE + SAMPLE\_OFF.$$

Here ϕ, λ and h denote the geodetic latitude, longitude, and height in the object space; LAT_OFF, $LONG\_OFF$, and $HEIGHT\_OFF$ are the offset values for the three ground coordinates, and LAT_SCALE, $LONG\_SCALE$, and $HEIGHT\_SCALE$ are the corresponding scale factors. Similarly p_x, p_y are the image coordinates, $LINE\_OFF$ and $SAMPLE\_OFF$ are the offset values for the two image coordinates, and $LINE\_SCALE$ and $SAMPLE\_SCALE$ are the scale factors for the two image coordinates. (Tong et al. [2])

In the implementation considered in this work, each polynomial is parameterized by twenty coefficients (third degree).

1.6 Ground Control Points (GCPs)

A Ground Control Point (GCP) is a point on the Earth's surface whose geographic location is known with high accuracy, typically through GPS or geodetic surveying measurements.

In addition to being accurately georeferenced, GCPs are usually selected so that they can be reliably identified in satellite imagery. Typical examples include road intersections, building corners, runway markings, or other visually distinctive features.

Given the coordinates of a GCP in the global frame, its position in the image can be predicted using either the PSM or the RFM. The localization error can then be measured as the distance between the predicted image location and the true image location of the GCP.

Numerous methods exist for determining the true image position of a GCP. The approach adopted in this work relies solely on the visual identification of the corresponding image feature. Although automating this process could potentially improve the accuracy and consistency of the obtained results, such methods are beyond the scope of this thesis. Their investigation is therefore left for future work.

Chapter 2

Model Correction

2.1 Physical Sensor Model

Let P_c and P_{in} denote the camera and intrinsic transformation matrices, respectively. Furthermore, let $\mathbf{s} = (s_x, s_y, s_z)$ be the sensed satellite position and $\mathbf{q} = (q_w, q_x, q_y, q_z)$ its sensed orientation quaternion expressed in the ECEF frame.

We assume that the camera calibration parameters and the intrinsic transformation matrix are sufficiently accurate and that any remaining errors are negligible compared to the errors introduced by the satellite position and orientation measurements. Consequently, the correction of the Physical Sensor Model can be restricted to correcting the satellite position and orientation.

Let

$$C : \mathbb{R}^n \times \mathbb{R}^7 \to \mathbb{R}^7 \tag{2.1}$$

be a correction function, where $\mathbb{R}^n$ denotes the parameter space. For example, a linear correction function may be represented by a 7×7 matrix and therefore requires $n = 49$ parameters.

Suppose that the errors affecting the satellite position and orientation can be approximated by the inverse of C for some parameter vector $\mathbf{P}$. In other words, data provided by satellite's sensors is distorted by C^{-1} for a given $\mathbf{P}$. Then corrected values (used to approximate the real position and orientation) can be obtained by applying

$$(\mathbf{s}_{\text{new}}, \mathbf{q}_{\text{new}}) = C(\mathbf{P}, \mathbf{s}, \mathbf{q}). \tag{2.2}$$

The choice of the correction function is largely arbitrary, as it only needs to approximate the systematic component of the error. However, determining suitable parameters requires the definition of a fitness function.

Let

$$F : \mathbb{R}^n \times \mathbb{R}^7 \times \mathbb{R}^3 \times \mathbb{R}^2 \to \mathbb{R} \tag{2.3}$$

be a fitness function used to evaluate the quality of the correction. The parameters of C can then be obtained by minimizing F using Ground Control Points (GCPs).

The correction procedure consists of the following steps:

1. Choose a correction function C. For example, a linear correction may be defined as

$$\mathrm{C}_{\mathbf{P}} = \begin{bmatrix} P_{11} & P_{12} & P_{13} & P_{14} & P_{15} & P_{16} & P_{17} \\ P_{21} & P_{22} & P_{23} & P_{24} & P_{25} & P_{26} & P_{27} \\ P_{31} & P_{32} & P_{33} & P_{34} & P_{35} & P_{36} & P_{37} \\ P_{41} & P_{42} & P_{43} & P_{44} & P_{45} & P_{46} & P_{47} \\ P_{51} & P_{52} & P_{53} & P_{54} & P_{55} & P_{56} & P_{57} \\ P_{61} & P_{62} & P_{63} & P_{64} & P_{65} & P_{66} & P_{67} \\ P_{71} & P_{72} & P_{73} & P_{74} & P_{75} & P_{76} & P_{77} \end{bmatrix},$$

   with

$$C(\mathbf{P}, \mathbf{s}, \mathbf{q}) = \mathrm{C}_{\mathbf{P}} \begin{bmatrix} \mathbf{s} \\ \mathbf{q} \end{bmatrix},$$

   where P_{ij} are the parameters to be optimized.

2. Choose a fitness function. One possible choice is

$$F(\mathbf{P}, \mathbf{s}, \mathbf{q}, x, y, z, p_x, p_y) = \mathrm{MSE}\Big(\mathrm{CM}(x, y, z), (p_x, p_y)\Big) + (1 - |\mathbf{q}_{\text{new}}|)^2,$$

   where MSE is a standard mean square error function, and CM denotes the Physical Sensor Model constructed using the corrected position and quaternion $(\mathbf{s}_{\text{new}}, \mathbf{q}_{\text{new}}) = C(\mathbf{P}, \mathbf{s}, \mathbf{q})$.

   The first term measures the projection error, while the second term enforces the unit-length constraint on the quaternion.

3. Optimize F in the parameter space using GCP coordinates (x, y, z) expressed in the ECEF frame and their corresponding image coordinates (p_x, p_y).

   For a given GCP, the values of $\mathbf{s}$ and $\mathbf{q}$ remain fixed during optimization. However different GCPs may rely on the same $\mathbf{s}$ and $\mathbf{q}$ if they are located in the same image.

4. To improve robustness, incorporate multiple GCPs into the optimization process. If

$$F_i : \mathbb{R}^n \ni \mathbf{P} \mapsto F(\mathbf{P}, \mathbf{s}_i, \mathbf{q}_i, x_i, y_i, z_i, p_{xi}, p_{yi}) \in \mathbb{R}$$

   denotes the fitness function associated with the i-th GCP, then the combined objective function can be defined as

$$F_{\text{total}} = \sum_i F_i.$$

2.2 Rational Function Model

For the Rational Function Model, the distinction between individual physical components of the imaging system is no longer available. Consequently, one may either optimize all RPCs (rational polynomial coefficients) or select a subset of coefficients according to a chosen heuristic, similarly to restricting optimization to the extrinsic parameters in the PSM case.

Assume that m coefficients have been selected for correction. Let

$$C : \mathbb{R}^n \times \mathbb{R}^m \to \mathbb{R}^m$$

be a correction function and

$$F : \mathbb{R}^n \times \mathbb{R}^m \times \mathbb{R}^3 \times \mathbb{R}^2 \to \mathbb{R}$$

a corresponding fitness function.

Unlike the PSM case, no additional quaternion normalization term is required. The optimization procedure is therefore slightly simpler.

1. Select the coefficients to be modified. For example, suppose that only the first two coefficients of D_x and the first three coefficients of N_y are chosen. Let the resulting coefficient vector be denoted by $\mathbf{c}$.

2. Choose a correction function. As an example, consider the quadratic model

$$C(\mathbf{P}, \mathbf{c}) = \mathbf{c}^T \mathrm{C}_q \mathbf{c} + \mathrm{C}_l \mathbf{c} + \mathrm{C}_b$$

where

$$\mathrm{C}_q = \begin{bmatrix} \mathbf{q}_{11} & \mathbf{q}_{12} & \mathbf{q}_{13} & \mathbf{q}_{14} & \mathbf{q}_{15} \\ \mathbf{q}_{21} & \mathbf{q}_{22} & \mathbf{q}_{23} & \mathbf{q}_{24} & \mathbf{q}_{25} \\ \mathbf{q}_{31} & \mathbf{q}_{32} & \mathbf{q}_{33} & \mathbf{q}_{34} & \mathbf{q}_{35} \\ \mathbf{q}_{41} & \mathbf{q}_{42} & \mathbf{q}_{43} & \mathbf{q}_{44} & \mathbf{q}_{45} \\ \mathbf{q}_{51} & \mathbf{q}_{52} & \mathbf{q}_{53} & \mathbf{q}_{54} & \mathbf{q}_{55} \end{bmatrix},$$

$$\mathrm{C}_l = \begin{bmatrix} l_{11} & l_{12} & l_{13} & l_{14} & l_{15} \\ l_{21} & l_{22} & l_{23} & l_{24} & l_{25} \\ l_{31} & l_{32} & l_{33} & l_{34} & l_{35} \\ l_{41} & l_{42} & l_{43} & l_{44} & l_{45} \\ l_{51} & l_{52} & l_{53} & l_{54} & l_{55} \end{bmatrix},$$

$$\mathrm{C}_b = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \\ b_5 \end{bmatrix}.$$

The parameter set is given by

$$\mathbf{P} = (q_{ijk}, l_{ij}, b_i).$$

3. Choose a fitness function, for example

$$F(\mathbf{P}, \mathbf{c}, B, L, H, p_x, p_y) = \mathrm{MSE}\Big(\mathrm{RF}(B, L, H), (p_x, p_y)\Big),$$

where RF denotes the Rational Function Model obtained after replacing the selected coefficients with their corrected values $\mathbf{c}_{\text{new}} = C(\mathbf{P}, \mathbf{c})$.

4. Optimize the fitness function using GCP geodetic coordinates and their corresponding image coordinates.

For a given GCP, the coefficient vector **c** remains constant. However, as in the case of PSM, different GCPs may rely on the same coefficient vector **c** if they are located in the same image.

5. As in the PSM case, multiple GCPs can be incorporated by defining

$$F_{\text{total}} = \sum_i F(\mathbf{P}, \mathbf{c}_i, x_i, y_i, z_i, p_{xi}, p_{yi}).$$

The source code is available on GitHub[1].

[1]Source code

Chapter 3

Results and Discussion

Priorly described correction approach for the Physical Sensor Model (PSM) and the Rational Function Model (RFM), was evaluated using Ground Control Points (GCPs) located in the San Francisco, California, and Cochabamba, Peru.

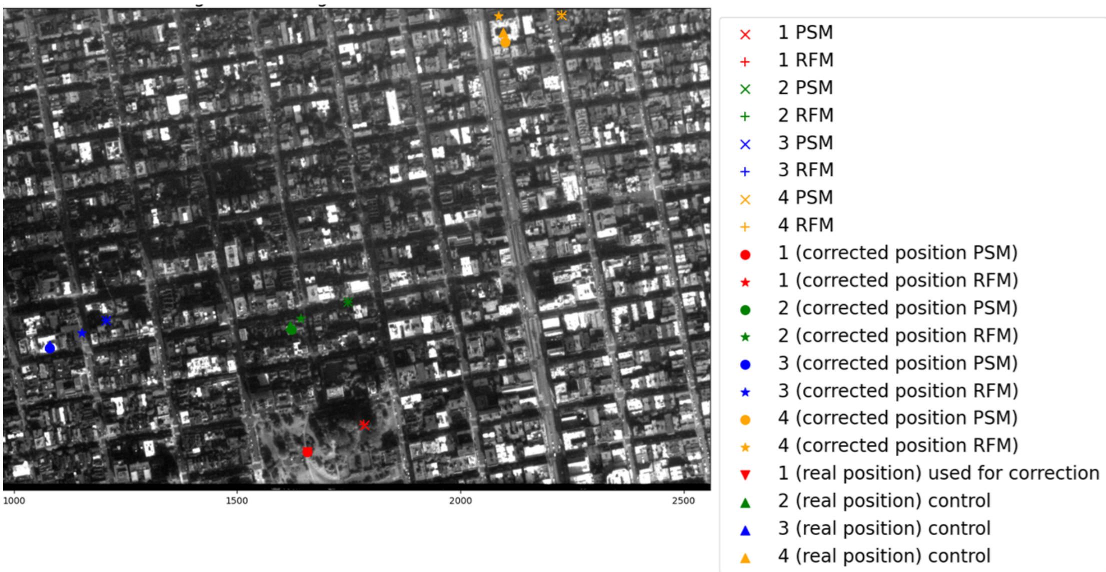

FIGURE 3.1: GCPs in San Francisco area together with their predicted and corrected position using PSM and RFM. Both models were corrected on a single GCP (red) using affine correction function

As illustrated in Figure 3.1, affine correction based on a single GCP (marked in red) can substantially improve the geopositioning accuracy of both the PSM and RFM models. The PSM has a potential advantage in this setting, as the correction is applied only to seven parameters describing the satellite position and orientation, which were assumed to be the primary sources of localization error. In contrast, correcting the RFM requires modifying all 80 coefficients, corresponding to the four third-degree polynomials comprising the model, each parameterized by 20 coefficients.

FIGURE 3.2: GCPs in San Francisco area together with their predicted and corrected position using PSM and RFM. Both models were corrected on five GCPs using quadratic correction function

Figure 3.2 demonstrates that, contrary to what might be expected, the use of a quadratic correction function does not necessarily lead to improved results. Intuitively, a model with a larger number of parameters offers greater flexibility and should therefore be capable of representing at least the same corrections as a simpler affine model. However, the increased model complexity also raises the risk of overfitting. In particular, the quadratic model may capture not only the systematic component of the localization error but also random noise, inaccuracies in the GCP measurements, and errors introduced during manual GCP identification. As a result, while the model may achieve a lower error on the training data, its performance on previously unseen control points can become less consistent and less predictable. (See Table 3.1.)

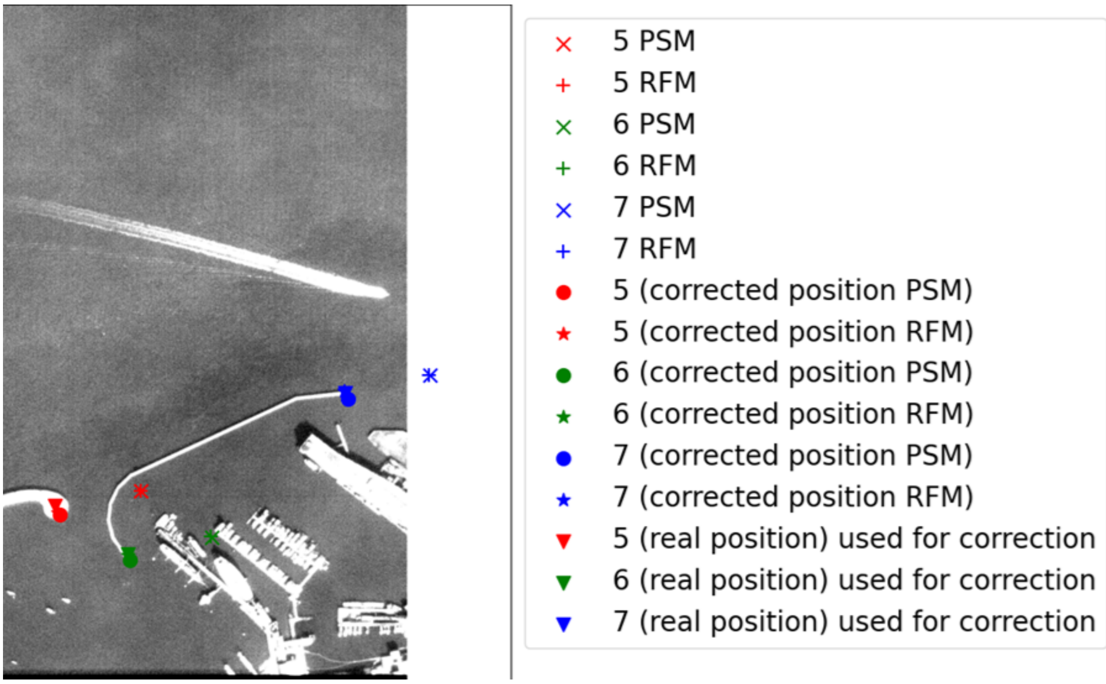

FIGURE 3.3: GCPs in San Francisco Bay area together with their predicted and corrected position using PSM and RFM. Both models were corrected on ten GCPs using shift correction function

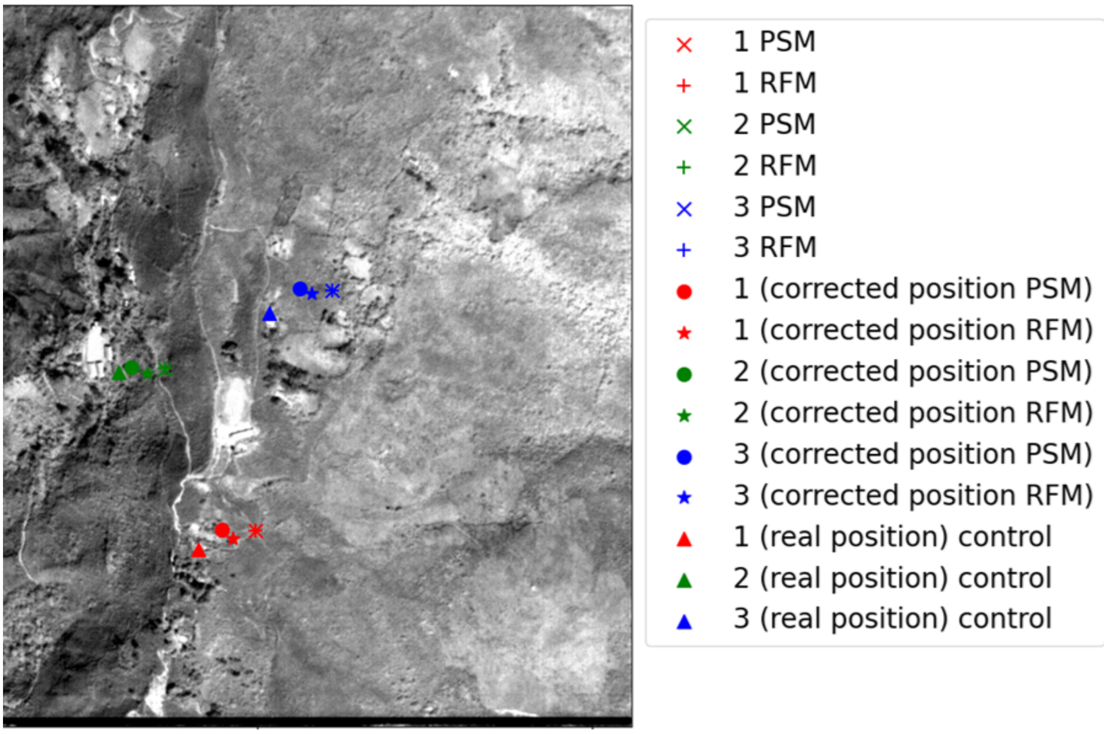

FIGURE 3.4: GCPs in Cochabamba, Peru together with their predicted and corrected position using PSM and RFM. Both models were corrected on a single GCP using affine correction function

Figure 3.4 demonstrates an improvement in geopositioning accuracy in Cochabamba obtained using a model corrected with a single GCP located in the San Francisco area.

This observation indicates that systematic errors in the sensed satellite position and orientation can be estimated in one geographic region and subsequently transferred to another. Therefore, the proposed correction method appears to capture platform-specific errors rather than location-dependent effects.

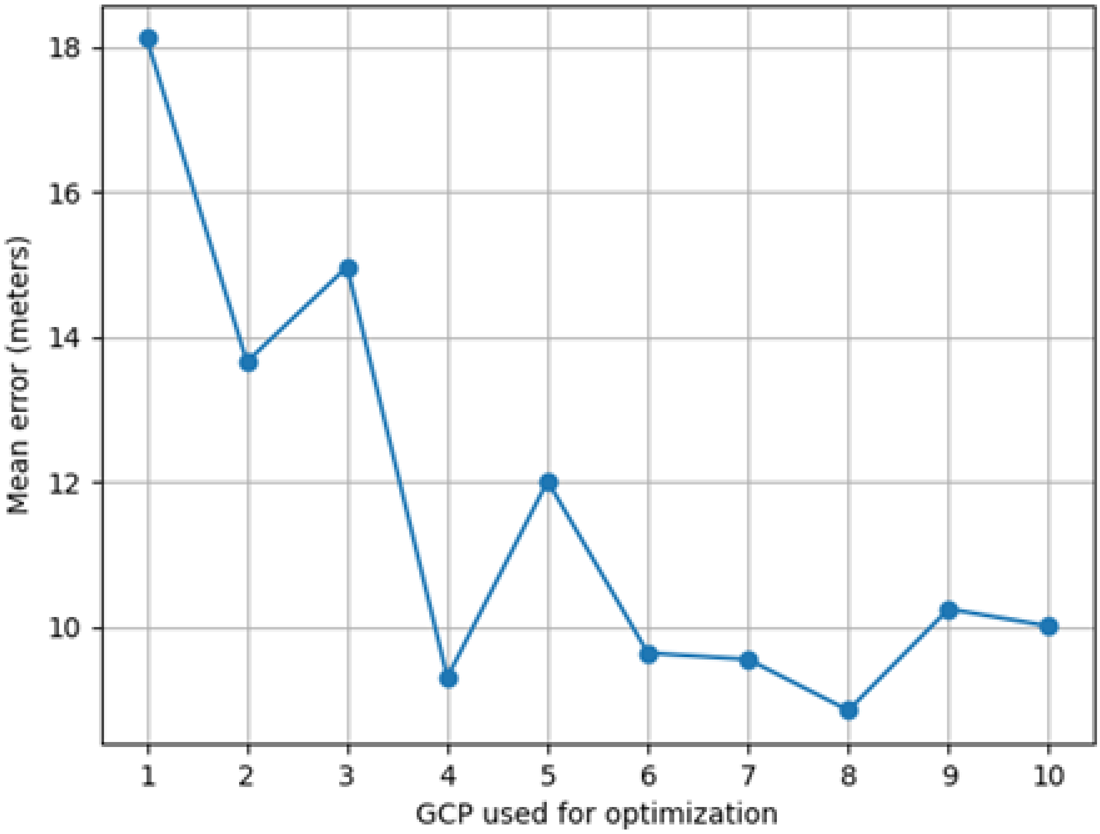

FIGURE 3.5: Graph showing difference in accuracy of PSM corrected using shift function based on the chosen GCP

Figure 3.5 provides evidence that the GCPs used in the experiments may have been affected by significant localization uncertainty. The figure shows that the accuracy measured on the control GCPs can vary by as much as 10 meters depending on which GCP was used for model correction. This level of variability suggests that the quality and positioning accuracy of the selected GCPs had a substantial impact on the obtained results.

Model	GCPs	Correction	SFT Accuracy [m]	SFC Accuracy [m]	CC Accuracy [m]
PSM	0	N/A	N/A	94.043037	59.817404
PSM	SF1	Shift	0.000001	18.116137	40.162478
PSM	SF1	Affine	0.022922	18.129745	28.861321
PSM	SF1	Quadratic	0.000000	18.107007	29.973553
PSM	SF5	Shift	9.698276	10.270634	44.928198
PSM	SF5	Affine	9.783020	10.428348	29.283147
PSM	SF5	Quadratic	9.749341	10.349308	32.331144
PSM	SF10	Shift	8.215481	N/A	49.077355
PSM	SF10	Affine	8.345980	N/A	29.583195
PSM	SF10	Quadratic	8.328720	N/A	32.833967

Table 3.1: Correction of the PSM using different datasets and correction functions

Model	GCPs	Correction	SFT Accuracy [m]	SFC Accuracy [m]	CC Accuracy [m]
FRM	0	N/A	N/A	93.808891	59.567831
FRM	SF1	Shift	0.000000	50.159627	32.747698
FRM	SF1	Affine	0.000000	51.555187	38.406227
FRM	SF1	Quadratic	0.037208	82.693869	38.339621
FRM	SF5	Shift	5.221465	13.688723	16.495237
FRM	SF5	Affine	0.241618	21.938688	18.126239
FRM	SF5	Quadratic	0.027991	19.227563	20.283893
FRM	SF10	Shift	5.458589	N/A	18.320117
FRM	SF10	Affine	2.461034	N/A	22.131488
FRM	SF10	Quadratic	0.721527	N/A	65.756774

Table 3.2: Correction of the RFM using different datasets and correction functions

The *GCPs* column of Tables 3.1 and 3.2 specifies the training dataset used to optimize the model. For example, *SF5* denotes a set of five GCPs located in the San Francisco area.

The *Correction* column indicates the type of correction function used during optimization.

The *SFT Accuracy* metric measures the average distance, expressed in meters, between the true and predicted locations of the GCPs used during training. Consequently, this metric reflects the model's performance on the training set.

The *SFC Accuracy* and *CC Accuracy* metrics measure the same quantity but are computed using independent control GCPs that were not used during optimization. The former is evaluated on control points located in the San Francisco area, while the latter is evaluated on control points located in the Cochabamba area.

The obtained results indicate that the Physical Sensor Model can be effectively improved by correcting only the satellite position and orientation parameters. This suggests that a substantial portion of the observed localization error may originate from inaccuracies in the satellite's navigation and attitude measurements rather than from the camera calibration itself.

Among the tested correction functions, the affine model achieved the best overall performance. However, this observation should be interpreted with caution. The experiment was conducted using a relatively small number of GCPs, and the quality of the available control points was limited. Consequently, it is difficult to determine whether the superior performance of the affine correction reflects the actual structure of the sensor errors or is merely a consequence of the characteristics of the experimental dataset.

Additional experiments involving a larger number of accurately surveyed GCPs and a wider range of geographic locations would be required to draw stronger conclusions regarding the optimal form of the correction function.

Appendix A

Representation of 3D rotations

To represent a rotation in the $\mathbb{R}^3$ vector space we can use many different conventions, however in this paper we will focus on the three most common ones - by angle and axis, by Euler's angles and by quaternions. We will start with a formal definition of a rotation, then introduce three ways of parameterizing rotations with their benefits and drawbacks, and at the and we will choose the method that suites the best our cause.

A.1 Rotations as a 3D Lie group

A.1.1 Definition of a rotation

Let's start this chapter with a rigorous definition of a rotation. Intuitively we know that the rotation shouldn't change the distance, angle or the angle's orientation between any two points. To put this formally we need two definitions:

Definition A.1. Let X, Y be the unitary linear spaces over $\mathbb{R}$ with dot products $\langle\cdot,\cdot\rangle_X$ and $\langle\cdot,\cdot\rangle_Y$ respectively. We say the mapping $f : X \to Y$ is **unitary** iff $\langle f(x_1), f(x_2)\rangle_Y = \langle x_1, x_2\rangle_X$ for every $x_1, x_2 \in X$.

This condition gives us a way to specify that a certain function preserves the distance and the angle, but not the angle's orientation as we can clearly see that the reflection through the XY plane is unitary but is not a rotation. So another definition is necessary:

Definition A.2. Let $n \in \mathbb{N}_1$, and $\mathbb{R}^n$ be a standard n-dimension linear space. We say that linear transformation $L : \mathbb{R}^n \to \mathbb{R}^n$ **preserves the orientation** iff $\det M(L) > 0$, where $M(L)$ is the matrix of L in the canonical base.

Now we can define what a rotation is:

Definition A.3. Rotation in $\mathbb{R}^3$, or **rotation** for short, is a linear mapping $L : \mathbb{R}^3 \to \mathbb{R}^3$, that is unitary and preserves the orientation.

It turns out that this definition lets us see the set of all rotations as a Lie group which is necessary for determining the dimension of this set. But before that let's look at three important properties of rotations.

A.1.2 Rotations' properties

Theorem A.4. *If R is a rotation then* $\det R = 1$.

Proof. Let R be any rotation and $x := (1, 0, 0)$, $y := (0, 1, 0)$, $z := (0, 0, 1)$. Using fundamental property of the determinant in $\mathbb{R}^n$ spaces we can write that $\mu_L(R([0,1]^3)) = |\det R| \cdot \mu_L([0,1]^3) = |\det R|$. We know that R is unitary so $(R(x), R(y), R(z))$ creates a unitary basis for $\mathbb{R}^n$ so it represents some cuboid. We also know that R preservers the norm so that cuboid is just a cube with an edge of length 1. Thus we can write that $\mu_L(R([0,1]^3)) = \mu_L([0,1]^3) = 1$, which with the previous equation gives us $|\det R| = 1$. Because R preserves the orientation $\det R = 1$.

□

Theorem A.5. *If R is a rotation, then it has at least one eigenvalue equal to 1.*

Proof. Let R be a rotation, $\lambda_1, \lambda_2, \lambda_3$ be its eigenvalues and v_1, v_2, v_3 its eigenvectors. Since $Rv_i = \lambda_i v_i$ and R preserves the norm we can write that $|v_i| = |Rv_i| = |\lambda_i v_i| = |\lambda_i||v_i|$. Because none of the eigenvectors can be zero we know that $|\lambda_i| = 1$. We also have that $1 = \det R = \lambda_1\lambda_2\lambda_3$ which is a determinant property. Now if all the eigenvalues are real, one or three of them are equal to 1. If two of them are complex then they are conjugate and their product is equal to one, thus making the third eigenvalue also equal to 1.

□

Theorem A.6. *If R is a rotation, then R^{-1} exists and $R^{-1} = R^T$.*

Proof. Let R be a rotation. By $\det R = 1 \neq 0$ we know that R^{-1} exists. Let's rewrite the unitary property of R as $(Rx)^T Ry = x^T y$. Then $x^T R^T Ry = x^T y$, and $R^T Ry = y$ thereby $R^T R = Id$.

□

A.1.3 Rotations as a group

Let's denote the set of rotations as $SO(3) := \{R \in \mathbb{R}^{3\times 3} \mid R \text{ is a rotation}\}$. Using this definition we obtain the following theorem:

Theorem A.7. *$SO(3)$ is a group with function composition as the group operation.*

Proof. Associativity and the existence of the identity element is given by the definition of the function composition. Let's prove that it is also closed. Let $R_1, R_2 \in SO(3)$, which means that $\det R_1 = \det R_2 = 1$. Using the property of the determinant we have that $\det R_1 \circ R_2 = \det R_1 \cdot \det R_2 = 1$, which proves the preservation of orientation. We also know that $\langle R_1(x_1), R_1(x_2)\rangle = \langle x_1, x_2\rangle$ and $\langle R_2(x_1), R_2(x_2)\rangle = \langle x_1, x_2\rangle$ for every $x_1, x_2 \in \mathbb{R}^3$. Let $y_1 := R_2(x_1),\ y_2 := R_2(x_2) \in \mathbb{R}^3$. We can now write that $\langle R_1 \circ R_2(x_1), R_1 \circ R_2(x_2)\rangle = \langle R_1(y_1), R_1(y_2)\rangle = \langle y_1, y_2\rangle = \langle R_2(x_1), R_2(x_2)\rangle = \langle x_1, x_2\rangle$, which proves the unitariness and thus that the operation is closed in $SO(3)$. Let's prove that for every rotation there exists the inverse rotation. Let $R \in SO(3)$. According to the theorem A.6. R^{-1} exists as the inverse function and $R^{-1} = R^T$. So $\det RR^{-1} = \det RR^T = \det R = 1$, which proves the preservation of orientation. We also know that $\forall_{x_1,x_2\in\mathbb{R}^3}\langle R(x_1), R(x_2)\rangle = \langle x_1, x_2\rangle$, which gives us $\forall_{x_1,x_2\in\mathbb{R}^3}\langle R(x_1), R(x_2)\rangle = \langle R^{-1}\circ R(x_1), R^{-1}\circ R(x_1)\rangle$, and together with the fact that R is surjective we have: $\forall_{y\in\mathbb{R}^3}\,\exists_{x\in\mathbb{R}^3}\, R(x) = y$ and finally $\forall_{y_1,y_2\in\mathbb{R}^3}\,\exists_{x_1,x_2\in\mathbb{R}^3}\, R(x_1) = y_1,\ R(x_2) = y_2 \wedge \langle y_1, y_2\rangle = \langle R^{-1}(y_1), R^{-1}(y_2)\rangle$, thus proving unitariness.

$\square$

A.1.4 $SO(3)$ as a manifold

It isn't easy to show that $SO(3)$ is a manifold and it's even harder to show that it's smooth or a Lie group. However intuitively it makes a lot of sense. First of all there are three axes around which we can rotate, and secondly using previously proved theorem A.6. we know that $R \cdot R^T = Id$ which results in 6 equations with 9 variables, thus leaving us with 3 independent variables. Because it's not relevant to the scope of this paper we will introduce some theorems without proofs.

Theorem A.8. *$SO(3)$ is a three-dimensional smooth manifold.*

Theorem A.9. *$\mathbb{D}^3_\pi \,/ \backsim$ is a three-dimensional smooth manifold.*

Theorem A.10. *$\mathbf{S}^3 \,/ \backsim$ is a three-dimensional smooth manifold.*

Theorem A.11. *$SO(3)$, $\mathbf{S}^3 \,/ \backsim$ and $\mathbb{D}^3_\pi \,/ \backsim$ are diffeomorphic.*

Lemma A.12. ***Euler's rotation theorem:***

For every rotation R there exists an axis $v \in \mathbb{R}^3$ and an angle $\alpha \in (-\pi, \pi]$ such that every affine plane A orthogonal to v is invariant and $R_{|A} : A \to A$ is a 2D rotation around v axis. In other words $R_{|A}(x) = L(L^{-1}(x)e^{i\alpha})$, where $L : \mathbb{R}^2 \to A \subset \mathbb{R}^3$ is an affine bijective transformation that maps 0 to the intersection of A and $\{\lambda v \mid \lambda \in \mathbb{R}\}$ and

$$e^{i\alpha} = \begin{pmatrix} \cos\alpha & \sin\alpha \\ -\sin\alpha & \cos\alpha \end{pmatrix} \tag{A.1}$$

Proof. Let R be any rotation and v be any eigenvector corresponding to eigenvalue 1. Let also L be a tangent vector space to v. Because R is unitary L has to be invariant. And indeed for every point $l \in L$ we have $0 = \langle l, v\rangle = \langle R(l), R(v)\rangle = \langle R(l), v\rangle$ and thus $R(l) \in L$. Because L is an invariant vector space we know that $R_{|L}$ is also unitary and preserves orientation so it is a 2D rotation and thus can be parametrized by a single angle α. Now let's consider a new base of $\mathbb{R}^3$ $\{v, u_1, u_2\}$ where u_1, u_2 are L base vectors. For every point $x \in \mathbb{R}^3$, and an affine plane A orthogonal to v containing x there exists $\lambda_1, \lambda_3, \lambda_3$ such that $x = \lambda_1 v + \lambda_2 u_1 + \lambda_3 u_2$. $R(x) = R(\lambda_1 v) + R(\lambda_2 u_1 + \lambda_3 u_2)$. Because $R(\lambda_2 u_1 + \lambda_3 u_2)$ belongs to L and $R(\lambda_1 v) = \lambda_1 v$ we know that A is also invariant and R can be treated as a 2D rotation around v axis on A thus proving the lemma. $\square$

A.2 Parametrization

A.2.1 Euler's angles

The most intuitive way to parametrize rotations is by using 3-dimensional torus $\mathbb{T}^3$ where every rotation can be described by three angles, each corresponding to the rotation around a canonical axis (X, Y or Z) called yaw, pitch and roll respectively. Then for every $a = (\alpha, \beta, \gamma) \in \mathbb{T}^3$ we can define rotation $R_a = R^z_\alpha \circ R^y_\beta \circ R^x_\gamma$, where

$$R^z_\alpha = \begin{pmatrix} \cos\alpha & \sin\alpha & 0 \\ -\sin\alpha & \cos\alpha & 0 \\ 0 & 0 & 1 \end{pmatrix} R^y_\alpha = \begin{pmatrix} \cos\alpha & 0 & \sin\alpha \\ 0 & 1 & 0 \\ -\sin\alpha & 0 & \cos\alpha \end{pmatrix} R^x_\alpha = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\alpha & \sin\alpha \\ 0 & -\sin\alpha & \cos\alpha \end{pmatrix} \tag{A.2}$$

It turns out that every rotation can be described by some three angles. But unfortunately this parametrization can't be a homeomorphism because it isn't an injection. Because of that lack of injectivity there exists a very famous problem with this parametrization called "gimbal lock". When the pitch angle reaches $90°$, other two angles become collinear, creating more ways to describe a certain rotation, which results in ambiguity. For example rotation of $(\frac{\pi}{2}, \frac{\pi}{2}, \frac{\pi}{2})$ is the same as the rotation of $(0, \frac{\pi}{2}, 0)$.

$$R = \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 0 \end{pmatrix}$$

There are also other problems like a trade-off between speed and precision when calculating sines and cosines, or inevitable numerical errors that make it hard for $R_\alpha \circ R_\beta \circ R_\gamma$ to remain unitary. There may also appear a problem of values "jumping" from $-\pi$ to π and vice-versa, which is a side effect of using quotient spaces. Because of all these reasons this method isn't ideal for our purposes.

A.2.2 Axis and angle

Second method relies on the Euler's rotation theorem that gives us a diffeomorphism between $\mathbb{D}^3_\pi / \backsim$ and $SO(3)$. If we define a rotation using a unit vector and an angle we can construct a parametrization R_x where $x \in \mathbb{D}^3_\pi / \backsim$ and R_x is a rotation around $\frac{x}{||x||}$ with an angle $||x||$. When $x = 0$, R_x is an identity. This way gimbal lock cannot occur. Unfortunately rotating a vector by some angle around some axis or creating the rotation matrix from that axis and angle are both very costly. There is also a problem when x approaches the boundary of the disk. We have to detect and fix it, moving it back inside and most likely reflecting it to the other side. So, even though this method seems much better, it still has some major issues.

A.2.3 Quaternions

The last parametrization relies on the diffeomorphism between $SO(3)$ and $\mathbf{S}^3 / \backsim$ which takes what's best from previous two methods. Now rotation R by a quaternion $q = (q_w, q_x, q_y, q_z) \in \mathbb{S}^3 / \backsim$ is defined as $(0, R(x, y, z)) = q(0, x, y, z)q^{-1}$. First of all there is no gimbal lock as it is also an injection. There are no "jumping" values from $-\pi$ to π and there's no need to deal with values "escaping" the sphere as in the case of the disc. We also reduced costly trig functions to simple multiplications. It seems that the only drawback is that quaternions have to stay normalized to work properly, but this condition is very easy to satisfy by just dividing the quaternion by its norm. Finally there is an easy way to construct the rotation matrix. Because of these and other reasons quaternions are so widely used in astrophysics and other fields as the go-to method for parameterizing rotations.

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